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Solution:

Python: Interview

1. What are Operators in Python?

Definition:
Operators are special symbols used to perform operations on variables and values.

Syntax:

operand1 operator operand2

Example:

```
a = 5
b = 3
print(a + b)  # Output: 8
```

Deep Explanation:
Here, + is an operator, and a, b are operands. Python uses operators to manipulate data like addition, comparison, logic building, etc.

2. What are the types of Operators in Python?

Definition:
Python provides different types of operators based on functionality.

Types:

1. Arithmetic
2. Comparison
3. Logical
4. Assignment
5. Bitwise
6. Membership
7. Identity

Example:

```
x = 10
print(x > 5)   # Comparison → True
print(x in [5, 10, 15])  # Membership → True
```

3. Explain Arithmetic Operators

Definition:
Used for mathematical calculations.

Operators: +, -, *, /, %, **, //

Syntax:

a operator b

Example:

```
a = 10
b = 3
print(a + b)   # 13
print(a % b)   # 1
print(a ** b)  # 1000
```

Deep Insight:

- / gives float result
- // gives floor division

4. Explain Comparison Operators

Definition:
Used to compare two values and return Boolean (True/False).

Operators: ==, !=, >, <, >=, <=

Example:

```
x = 5
y = 10
print(x < y)   # True
print(x == y)  # False
```

5. What are Logical Operators?

Definition:
Used to combine conditional statements.

Operators: and, or, not

Syntax:

condition1 and condition2

Example:

```
a = 5
b = 10
print(a > 2 and b > 5)  # True
print(not(a > 2))       # False
```

Deep Insight:

- and → both true
- or → at least one true
- not → reverse result

6. Explain Assignment Operators

Definition:
Used to assign values to variables.

Operators: =, +=, -=, *=, /=, //=, %=, **=

Example:

```
x = 5
x += 3   # same as x = x + 3
print(x) # 8
```

7. What are Bitwise Operators?

Definition:
Used to perform operations on binary numbers.

Operators: &, |, ^, ~, <<, >>

Example:

```
a = 5   # 101
b = 3   # 011
print(a & b)  # 1 (001)
```

Deep Insight:
Works at bit level → used in low-level programming.

8. Explain Membership Operators

Definition:
Used to check if a value exists in a sequence.

Operators: in, not in

Example:

```
nums = [1, 2, 3, 4]
print(2 in nums)      # True
print(5 not in nums)  # True
```

9. Explain Identity Operators

Definition:
Used to compare memory location of two objects.

Operators: is, is not

Example:

```
a = [1, 2]
b = a
print(a is b)  # True
```

Deep Insight:

- == → compares value
- is → compares memory

10. Operator Precedence in Python

Definition:
Determines the order in which operations are evaluated.

Example:

```
result = 10 + 2 * 3
print(result)  # 16
```

Explanation:

- * has higher precedence than +

Order (High → Low):

1. **
2. *, /, //, %
3. +, -
4. Comparison
5. Logical